Target-Weighted Bellman Aggregation Trees and Forests for Offline Policy Evaluation

Anonymous Author(s)
Affiliation
Address
email

Abstract

Regression trees and random forests are natural tools for interpretable and adaptive value estimation, but tree-based fitted value methods usually treat Bellman updates as ordinary supervised labels. The resulting splits are optimized for transient pseudo-outcomes and for the sampling distribution of the offline data, which need not match the distribution under which value accuracy matters. We propose *target-weighted Bellman Aggregation Trees and Forests* for offline policy evaluation. A tree defines a partition, while a forest defines a finite collection of leaf-indicator features. In both cases, the final value estimate is obtained by solving one target-weighted projected Bellman equation, rather than by iterating supervised regressions. Single Bellman Aggregation Trees (BATs) give the interpretable partition special case; Bellman Aggregation Forests (BAFs) use randomized honest trees to build a richer finite feature space before a joint Bellman solve. For fixed or honest learned features, the population estimator is the projected Bellman fixed point, with approximation error controlled by the projection error of Q^π. Finite samples, rank reduction, validation pruning, and estimated weights enter as perturbations of the projected Bellman linear system. The manuscript gives a method and theory draft with a falsifiable experimental protocol; empirical BAT/BAF results remain pending.

1 Introduction

Tree-based value estimation is appealing for a simple reason: a tree gives a readable partition of the state-action space. If a fitted value function assigns the same number to all points in a leaf, one would like that leaf to mean something dynamical: points in the same leaf should have similar immediate rewards and similar continuation behavior under the policy being evaluated, in the distribution where accuracy matters. Standard tree-based fitted value methods do not enforce this interpretation. They usually embed regression trees or tree ensembles inside fitted Bellman iteration: at each round, a Bellman target is computed from the previous value estimate and a supervised regressor is fit to that target [6, 10]. This is a useful modular algorithm, but the fitted tree is optimized for a moving pseudo-label, usually under the behavior distribution. A split may reduce prediction error for the current target while merging states whose target-policy transitions lead to different parts of the value function.

This paper starts from the opposite perspective. A tree is not only a supervised learner; it is also a finite aggregation of the state-action space. A random forest is not only an average of supervised trees; it is also a finite, adaptive feature map built from many leaf indicators. Once such a feature map and an evaluation distribution μ are fixed, policy evaluation has a canonical solution: project the Bellman equation onto the span of those features under μ. For a single tree this projection is exactly the aggregate Bellman equation between leaves. For a forest it is a single linear projected Bellman solve over all forest leaf features.

Submitted to 40th Conference on Neural Information Processing Systems (NeurIPS 2026). Do not distribute.

We propose *target-weighted Bellman Aggregation Trees and Forests*. Given offline transitions (S_i, A_i, R_i, S_i') from a behavior distribution ν, the method learns tree or forest features over $X = (S, A)$, evaluates them under a reference distribution μ, typically the target policy's stationary state-action distribution, and uses weights $w = d\mu/d\nu$ to fit the projected Bellman system. Splits are chosen by weighted held-out Bellman residuals computed after provisional Bellman solves. The final value estimate is then refit on an independent estimation sample. For BAF, all provisional per-tree values are discarded; the retained object is the forest feature map, and the final values come from one joint projected Bellman solve.

The contribution is deliberately narrow. BATs and BAFs are not proposed as deep offline RL systems. They address a more basic statistical question: how should one fit regression trees and random forests when the regression target is a Bellman fixed point and the evaluation distribution differs from the sampling distribution? The answer is to treat tree leaves as Bellman aggregation features, then solve the target-weighted projected Bellman equation induced by those features.

Our contributions are:

- We formulate BATs and BAFs as non-iterative estimators for offline policy evaluation: a single tree is a partition-indicator feature map, and a forest is a randomized collection of leaf-indicator features.
- We give a practical construction using weighted Bellman-aware split scores, honest sample splitting, stabilized weights, validation pruning, and honest rank reduction before the final forest solve.
- We prove that, for fixed or honest learned features, the estimator solves a projected Bellman equation and has aggregation bias controlled by $\|(I - \Pi_\psi^\mu)Q^\pi\|_{L^2(\mu)}$. For trees this is within-leaf variation; for forests it is projection error onto the forest leaf-feature span.
- We state finite-sample and weight-estimation perturbation bounds for the projected Bellman linear system, plus a finite-candidate validation guarantee.
- We give an experimental protocol, with results pending, designed to isolate whether Bellman-aware tree and forest features improve value accuracy per degree of freedom compared with ordinary supervised tree objectives.

2 Problem Setup

We consider an infinite-horizon discounted Markov decision process with state space $\mathcal{S}$, finite action space $\mathcal{A}$, transition kernel K, bounded reward $|R| \le R_{\max}$, and discount $\gamma \in (0,1)$. The target policy $\pi(a \mid s)$ is fixed. Let $X = (S, A) \in \mathcal{X} = \mathcal{S} \times \mathcal{A}$, and define

$$X_+^\pi = (S', A'), \qquad S' \sim K(\cdot \mid S, A), \quad A' \sim \pi(\cdot \mid S').$$

The policy-evaluation Bellman operator is

$$(T^\pi Q)(s,a) = \mathbb{E}[R + \gamma Q(S', A') \mid S = s, A = a,\ A' \sim \pi(\cdot \mid S')],$$

and Q^π is its unique fixed point. We observe n i.i.d. one-step offline transitions (S_i, A_i, R_i, S_i') with $X_i = (S_i, A_i) \sim \nu$. Trajectory data require standard mixing or blocking arguments. We evaluate error under a reference distribution μ on $\mathcal{X}$, typically the target-policy stationary state-action distribution μ_π, and assume $\mu \ll \nu$ with density ratio $w = d\mu/d\nu$.

Finite Bellman features. Let $\psi : \mathcal{X} \to \mathbb{R}^d$ be a finite feature map selected independently of the final estimation sample. It represents value functions

$$Q_\theta(x) = \psi(x)^\top \theta, \qquad \theta \in \mathbb{R}^d.$$

For a next state s', write

$$\bar{\psi}(s') = \mathbb{E}_{A' \sim \pi(\cdot|s')}[\psi(s', A')].$$

The target-weighted projected Bellman equations are

$$A_\psi^\mu \theta = b_\psi^\mu, \qquad A_\psi^\mu = \mathbb{E}_\mu\left[\psi(X)\{\psi(X) - \gamma\psi(X_+^\pi)\}^\top\right], \quad b_\psi^\mu = \mathbb{E}_\mu[\psi(X)R].$$

With behavior data, these moments are estimated by importance weighting:

$$\widehat{A}_\psi = \frac{\sum_i \hat{w}_i \psi_i (\psi_i - \gamma \bar{\psi}_i^+)^\top}{\sum_i \hat{w}_i}, \qquad \widehat{b}_\psi = \frac{\sum_i \hat{w}_i \psi_i R_i}{\sum_i \hat{w}_i},$$

where $\psi_i = \psi(S_i, A_i)$ and $\bar{\psi}_i^+ = \bar{\psi}(S_i')$. The final estimate is $\hat{Q}_\psi(x) = \psi(x)^\top \hat{\theta}$, where $\widehat{A}_\psi \hat{\theta} = \widehat{b}_\psi$ after deterministic rank reduction if needed.

Trees and forests as features. A regression tree $\mathcal{T}$ partitions $\mathcal{X}$ into leaves $\ell \in \{1, \dots, L\}$. Its feature map is $\psi_\mathcal{T}(x) = e_{\phi_\mathcal{T}(x)}$, the one-hot indicator of the leaf containing x. In this case (2) is exactly the aggregate Bellman equation

$$q = r^\mu + \gamma P^\mu q, \qquad r_\ell^\mu = \mathbb{E}_\mu[R \mid X \in \ell], \quad P_{\ell m}^\mu = \Pr_\mu\{\phi_\mathcal{T}(X_+^\pi) = m \mid X \in \ell\}.$$

A random forest $\mathcal{F} = (\mathcal{T}_1, \dots, \mathcal{T}_B)$ defines the concatenated leaf feature map

$$\psi_\mathcal{F}(x) = \frac{1}{B}\left(e_{\phi_{\mathcal{T}_1}(x)}^\top, \dots, e_{\phi_{\mathcal{T}_B}(x)}^\top\right)^\top.$$

The scaling is fixed and immaterial after rank reduction; it keeps the forest features on a comparable numerical scale as B changes. BAF solves (2) once over the span of these forest features. It is not the average of B independently labeled BATs.

3 Bellman Aggregation Trees and Forests

The method has two stages. First, it constructs tree or forest features using Bellman-aware split scores. Second, it discards provisional split-time values and solves the final target-weighted projected Bellman system on an independent estimation sample. The theorem-clean version uses five disjoint sample roles: growth-fit, growth-score, validation, rank, and estimation. Cross-fitting can recover data efficiency, but honesty keeps the estimand transparent.

3.1 Bellman-Aware Split Scoring

For a candidate feature map ψ, estimate $\widehat{A}_\psi, \widehat{b}_\psi$ on the growth-fit sample and solve $\widehat{A}_\psi \hat{\theta}_\psi = \widehat{b}_\psi$. Score the candidate on a held-out sample by

$$\delta_i(\psi) = R_i + \gamma(\bar{\psi}_i^+)^\top \hat{\theta}_\psi - \psi_i^\top \hat{\theta}_\psi, \qquad \mathrm{Score}_\mu(\psi) = \frac{\sum_{i \in \mathcal{D}_{\mathrm{hold}}} \hat{w}_i \delta_i(\psi)^2}{\sum_{i \in \mathcal{D}_{\mathrm{hold}}} \hat{w}_i} + \lambda \dim(\operatorname{span} \psi).$$

During growth, $\mathcal{D}_{\mathrm{hold}}$ is the growth-score sample. The validation sample is used only to prune a finite path or finite forest candidate set.

For a single BAT, the candidate maps are one-hot leaf indicators for candidate tree splits, and the provisional solve can be implemented as the aggregate leaf Bellman solve. For BAF, randomized BAT-style trees are grown using subsampling and feature subsampling. The split score is a feature-construction criterion; the final forest values are obtained only after all trees have been grown and the joint forest feature map has been rank reduced.

3.2 Stabilized Weights and Rank Reduction

Given raw density-ratio estimates $\tilde{w}_i$, the practical estimator clips and shrinks them before tree construction:

$$\bar{w}_i = (1-\alpha)\min\{\tilde{w}_i, c\} + \alpha, \qquad \hat{w}_i = \frac{\bar{w}_i}{|\mathcal{D}_s|^{-1} \sum_{j \in \mathcal{D}_s} \bar{w}_j}, \quad i \in \mathcal{D}_s.$$

Here $\mathcal{D}_s$ is one of the sample roles. A leaf is admissible only if it has enough raw observations, enough weighted mass, and effective sample size

$$\mathrm{ESS}(\ell) = \frac{\left(\sum_{i:\phi(X_i)=\ell} \hat{w}_i\right)^2}{\sum_{i:\phi(X_i)=\ell} \hat{w}_i^2}$$

Algorithm 1 Target-Weighted Bellman Aggregation Forest

Require: Offline data $\mathcal{D}$, target policy π, raw weights $\tilde{w}$, number of trees B, discount γ, split and pruning parameters, stabilization parameters.

```
Split D into growth-fit, growth-score, validation, rank, and estimation samples.
Stabilize and normalize w̃ within each sample to obtain ŵ.
for b = 1, ..., B do
    Draw randomized growth-fit and growth-score subsamples and feature subsets.
    Grow a BAT-style tree path using admissible splits with minimum raw count, weighted mass,
    and ESS.
    Score candidate splits by (3.1) using provisional Bellman solves on the current tree.
    Prune the finite path on the validation sample and store the selected tree T_b.
end for
Build ψ_F from all selected forest leaves; for BAT take B = 1.
Drop linearly dependent columns by deterministic QR/SVD on the rank sample.
Estimate Â_{ψ_F} and b̂_{ψ_F} by (2).
Solve Â_{ψ_F} θ̂ = b̂_{ψ_F}.
return Q̂(x) = ψ_F(x)^T θ̂.
```

above a prespecified threshold. Forest features are often linearly dependent: each tree contributes leaf indicators that sum to a constant, and different trees may create redundant columns. The theorem-clean implementation performs deterministic rank reduction on an independent rank sample before the projected Bellman solve. Using the estimation design for QR/SVD is a pragmatic implementation shortcut because it uses no rewards, but the stated finite-sample theorem treats the retained full-rank feature map as fixed before final estimation. Ridge regularization is reported separately because it changes the estimand.

4 Theory

The theory treats a learned tree or forest as fixed after conditioning on the growth, validation, and rank-reduction data. The central object is the feature span $\mathcal{V}_\psi = \{\psi^\top\theta : \theta \in \mathbb{R}^d\}$, not the particular algorithm that produced it. After rank reduction, $d = d_r = \dim(\mathcal{V}_\psi)$ denotes the retained feature dimension. We write

$$(P_\pi f)(x) = \mathbb{E}[f(X_+^\pi) \mid X = x], \qquad \Pi_\psi^\mu f = \arg\min_{g \in \mathcal{V}_\psi} \|f - g\|_{L^2(\mu)}$$

for the target-policy transition operator and the $L^2(\mu)$ orthogonal projection onto the retained feature span.

Assumption 1 (Sampling, overlap, and boundedness)**.** The final estimation sample consists of i.i.d. one-step transitions with $X_i \sim \nu$. The reference measure satisfies $\mu \ll \nu$, the oracle weight $w = d\mu/d\nu$ is known for the main finite-sample theorem and bounded by W, rewards satisfy $|R| \le R_{\max}$, and $\gamma \in (0, 1)$.

Assumption 2 (Reference-measure contraction)**.** The Bellman operator is a γ-contraction in $L^2(\mu)$:

$$\|T^\pi f - T^\pi g\|_{L^2(\mu)} \le \gamma \|f - g\|_{L^2(\mu)} \qquad \text{for all bounded } f, g.$$

This holds, for example, when $\mu = \mu_\pi$ is a stationary state-action distribution for the target-policy Markov chain.

Lemma 1 (Stationary reference measures are contraction measures)**.** *If μ_π is stationary for the target-policy Markov chain on $\mathcal{X}$, then T^π is a γ-contraction in $L^2(\mu_\pi)$.*

Assumption 3 (Honest finite features and rank)**.** The feature map $\psi : \mathcal{X} \to \mathbb{R}^d$ is selected independently of the final estimation sample, satisfies $\|\psi(x)\|_2 \le K$, and has full rank after deterministic rank reduction. The population projected Bellman matrix A_ψ^μ in (2) has $\sigma_{\min}(A_\psi^\mu) \ge \kappa > 0$.

Proposition 1 (Projected Bellman target)**.** *Under Assumption 2, the operator $\Pi_\psi^\mu T^\pi$, restricted to $\mathcal{V}_\psi$, is a γ-contraction, where Π_ψ^μ is the $L^2(\mu)$ projection onto $\mathcal{V}_\psi$. Hence there is a unique $Q_\psi^\mu \in \mathcal{V}_\psi$ satisfying*

$$Q_\psi^\mu = \Pi_\psi^\mu T^\pi Q_\psi^\mu.$$

If A_ψ^μ is nonsingular, then $Q_\psi^\mu(x) = \psi(x)^\top \theta_\psi^\mu$ with $A_\psi^\mu \theta_\psi^\mu = b_\psi^\mu$.

Theorem 1 (Aggregation bias for trees and forests)**.** *Under Assumption 2, any fixed tree or forest feature map ψ satisfies*

$$\|Q_\psi^\mu - Q^\pi\|_{L^2(\mu)} \le \frac{1}{1-\gamma} \|(I - \Pi_\psi^\mu) Q^\pi\|_{L^2(\mu)}.$$

For a BAT, let $\Pi_\mathcal{T}^\mu$ denote the $L^2(\mu)$ projection onto the tree leaves. The right-hand side is the within-leaf variation of Q^π under μ:

$$\|(I - \Pi_\mathcal{T}^\mu) Q^\pi\|_{L^2(\mu)}^2 = \mathbb{E}_\mu[\mathrm{Var}\{Q^\pi(X) \mid \phi_\mathcal{T}(X)\}].$$

For a BAF, it is the approximation error of Q^π by the span of all retained forest leaf features.

Proposition 2 (Forest feature monotonicity)**.** *If $\mathcal{V}_{\psi_1} \subseteq \mathcal{V}_{\psi_2}$, then*

$$\|(I - \Pi_{\psi_2}^\mu) Q^\pi\|_{L^2(\mu)} \le \|(I - \Pi_{\psi_1}^\mu) Q^\pi\|_{L^2(\mu)}.$$

Thus, after rank reduction, adding linearly independent forest leaf features cannot increase the population approximation term, although the finite-sample error depends on the resulting dimension and conditioning of A_ψ^μ.

Theorem 2 (Projected Bellman perturbation)**.** *Let $\theta = \theta_\psi^\mu$ and $\hat\theta$ solve $A_\psi^\mu \theta = b_\psi^\mu$ and $\widehat{A}_\psi \hat\theta = \widehat{b}_\psi$. If*

$$\|\widehat{A}_\psi - A_\psi^\mu\|_{\mathrm{op}} \le \varepsilon_A < \kappa, \qquad \|\widehat{b}_\psi - b_\psi^\mu\|_2 \le \varepsilon_b,$$

then

$$\|\hat\theta - \theta\|_2 \le \frac{\varepsilon_b + \varepsilon_A \|\theta\|_2}{\kappa - \varepsilon_A}, \qquad \|\hat{Q}_\psi - Q_\psi^\mu\|_{L^2(\mu)} \le K \frac{\varepsilon_b + \varepsilon_A \|\theta\|_2}{\kappa - \varepsilon_A}.$$

Proposition 3 (A simple concentration consequence)**.** *Assume Assumptions 1 and 3, with oracle weights on the final estimation sample. Let*

$$D_n = \frac{1}{n}\sum_{i=1}^n w_i, \qquad \widetilde{A}_\psi = \frac{1}{n}\sum_{i=1}^n w_i \psi_i (\psi_i - \gamma \bar\psi_i^+)^\top, \qquad \widetilde{b}_\psi = \frac{1}{n}\sum_{i=1}^n w_i \psi_i R_i,$$

and $\widehat{A}_\psi = \widetilde{A}_\psi / D_n$, $\widehat{b}_\psi = \widetilde{b}_\psi / D_n$. There is a universal constant C such that, if $n \ge 2W^2 \log(6/\delta)$, then with probability at least $1-\delta$,

$$\|\widehat{A}_\psi - A_\psi^\mu\|_{\mathrm{op}} \le CWK^2(1+\gamma)\left\{\sqrt{\frac{d + \log(6/\delta)}{n}} + \frac{\log(6d/\delta)}{n}\right\},$$

$$\|\widehat{b}_\psi - b_\psi^\mu\|_2 \le CWKR_{\max}\sqrt{\frac{d + \log(6/\delta)}{n}}.$$

This elementary bound is not intended to be sharp; it makes explicit the forest tradeoff between feature dimension, conditioning, and sample size.

Corollary 1 (Tree aggregate special case)**.** *For one-hot tree features, $A_\psi^\mu = D_\mu(I - \gamma P^\mu)$ and $b_\psi^\mu = D_\mu r^\mu$, where $D_\mu = \mathrm{diag}(\mu_1, \dots, \mu_L)$, $\mu_\ell = \mu\{\phi_\mathcal{T}(X) = \ell\}$, and $\mu_{\min} = \min_\ell \mu_\ell > 0$. Let*

$$\hat\mu_\ell = \frac{1}{n}\sum_{i=1}^n w_i \mathbb{1}\{\phi_\mathcal{T}(X_i) = \ell\}.$$

If $\min_\ell \hat\mu_\ell \ge \mu_{\min}/2$, and $\|\widehat{r}^\mu - r^\mu\|_\infty \le \varepsilon_r$, $\|\widehat{P}^\mu - P^\mu\|_\infty \le \varepsilon_P$, then

$$\|\widehat{q}_\mathcal{T}^\mu - q_\mathcal{T}^\mu\|_\infty \le \frac{\varepsilon_r}{1-\gamma} + \frac{\gamma R_{\max} \varepsilon_P}{(1-\gamma)^2}.$$

The leaf-mass condition holds with probability at least $1-\delta$ when

$$n \ge 2W^2 \mu_{\min}^{-2} \log(2L/\delta).$$

Proposition 4 (Estimated-weight perturbation)**.** *Let θ^w and $\theta^{\hat{w}}$ be the projected Bellman solutions obtained from two weighted moment systems (A^w, b^w) and $(A^{\hat{w}}, b^{\hat{w}})$. If $\sigma_{\min}(A^w) \geq \kappa$, $\|A^{\hat{w}} - A^w\|_{\mathrm{op}} \leq \eta_A < \kappa$, and $\|b^{\hat{w}} - b^w\|_2 \leq \eta_b$, then*

$$\|\theta^{\hat{w}} - \theta^w\|_2 \leq \frac{\eta_b + \eta_A \|\theta^w\|_2}{\kappa - \eta_A}.$$

Thus estimated or stabilized weights affect BAT/BAF only through the aggregate projected Bellman moments they induce.

Proposition 5 (Validation pruning over a finite candidate set)**.** *Let $\mathfrak{P}$ be a finite set of candidate trees, forests, or feature maps selected independently of a validation sample of size n_{val}. For each $\psi \in \mathfrak{P}$, let δ_ψ be its validation residual after the provisional Bellman solve. Suppose $|\delta_\psi| \leq B_\delta$ and $0 \leq w \leq W$. If $\hat{\psi}$ minimizes the empirical weighted residual plus $\lambda \dim(\mathrm{span}\, \psi)$ over $\mathfrak{P}$, then with probability at least $1 - \delta$,*

$$L_\mu(\hat{\psi}) + \lambda \dim(\mathrm{span}\, \hat{\psi}) \leq \min_{\psi \in \mathfrak{P}} \{L_\mu(\psi) + \lambda \dim(\mathrm{span}\, \psi)\} + 2WB_\delta^2 \sqrt{\frac{\log(2|\mathfrak{P}|/\delta)}{2n_{\mathrm{val}}}},$$

where $L_\mu(\psi) = \mathbb{E}_\mu[\delta_\psi^2]$. Here the empirical risk is the unnormalized oracle-weight risk

$$\widehat{L}_{\mathrm{val}}(\psi) = \frac{1}{n_{\mathrm{val}}} \sum_{i \in \mathcal{D}_{\mathrm{val}}} w_i \delta_{\psi,i}^2.$$

The self-normalized validation score obeys the same comparison up to the standard denominator event for $n_{\mathrm{val}}^{-1} \sum_{i \in \mathcal{D}_{\mathrm{val}}} w_i$.

Combining the population and sample terms gives

$$\|\hat{Q}_\psi - Q^\pi\|_{L^2(\mu)} \leq \underbrace{\|Q_\psi^\mu - Q^\pi\|_{L^2(\mu)}}_{\text{projection bias}} + \underbrace{\|\hat{Q}_\psi - Q_\psi^\mu\|_{L^2(\mu)}}_{\text{statistical error}}.$$

Honesty lets this bound be read conditionally on the learned tree or forest.

5 Experimental Protocol

The empirical part of this manuscript remains a protocol rather than a completed results section. The protocol tests the mechanism: whether target-weighted Bellman-aware tree and forest features improve value accuracy per degree of freedom when supervised prediction boundaries and Bellman-relevant transition boundaries differ.

Diagnostic MDP. Use a small continuous-state MDP where immediate reward variation and continuation-value variation occur along different axes. A supervised tree or forest fit to Monte Carlo returns or one-step pseudo-labels can split on reward while merging states with different target-policy continuations. The diagnostic figure reports the true Q^π, learned BAT partitions, forest-induced nearest-neighbor/feature weights, and scalar errors.

Grid-auditable navigation. Use a two-dimensional navigation task with discrete actions, nonlinear drift, and offline data whose behavior distribution under-samples regions important under μ_π. A fine grid solver supplies reference values and stationary norm diagnostics; the learning algorithms see only offline transitions.

Baselines and ablations. Compare BAT and BAF against CART on Monte Carlo returns, random-forest returns, Tree-FQE, RF-FQE, random tree partitions, and uniform grids. The main ablation compares supervised split scores with Bellman-aware split scores. A second ablation compares the joint BAF solve with bagged independently solved BATs, which tests whether the projected forest formulation matters.

Metrics. Report stationary $L^2(\mu_\pi)$ value error, weighted Bellman residual, policy-value error, error versus number of leaves or retained features, and error versus number of trees. The target pattern is lower value error per degree of freedom and partitions or forest weights aligned with transition-induced value boundaries. Residuals are diagnostic, not a substitute for value error under μ_π.

6 Related Work

Tree-based fitted value methods. Tree-based batch-mode reinforcement learning uses regression trees and tree ensembles inside fitted value iteration [6]. The policy-evaluation analogue, Tree-FQE or RF-FQE, treats trees as supervised regressors for Bellman pseudo-labels. BATs and BAFs instead use trees to build aggregation features and compute values by a projected Bellman solve.

Random forests and forest features. Random forests average randomized trees to obtain adaptive nonparametric regressors [4]. Modern work also views forests through adaptive kernels, feature spans, and honesty [1, 11, 16]. BAF uses the forest as an adaptive Bellman feature map: the final estimate is not a supervised forest average, but the projected fixed point over all retained leaf features.

State aggregation and projected Bellman equations. State aggregation reduces dynamic programming to a finite abstract process [3, 15]. Least-squares temporal difference methods solve projected Bellman equations under linear function approximation [9, 13]. BAT is the tree-partition instance of this idea; BAF is the random-forest feature instance.

Bellman residual and minimax methods. Bellman residual minimization has a long history [2]. Modern offline RL methods use minimax or adversarial Bellman-error objectives to avoid strong completeness assumptions [14, 18, 19]. BAT/BAF use residuals in a restricted way: residuals construct and prune features, while final values come from the projected Bellman equation.

7 Discussion and Limitations

BATs and BAFs answer a narrow question: how should tree and forest regressors be fit when the target is a Bellman fixed point and the evaluation distribution differs from the sampling distribution? The proposed answer is to use trees as target-weighted aggregation features and solve the induced projected Bellman equation. Single trees retain an interpretable partition; forests trade some of that direct interpretability for a richer approximation space.

The paper focuses on fixed-policy evaluation. For optimal control, the Bellman equation contains a max over actions, so the final linear solve no longer applies. A second limitation is conditioning: rich forest feature spaces can reduce approximation bias while increasing variance and numerical ill-conditioning, which is why deterministic rank reduction is part of the estimator. Finally, this manuscript is evidence-incomplete. It specifies the experiments needed to test the mechanism, but does not yet report BAT/BAF results.

Reproducibility Statement

The implementation uses deterministic data splits, including a rank-reduction split for the theorem-clean version, recorded random seeds for forest subsampling and feature subsampling, declared weight stabilization, minimum raw-count, weighted-mass and ESS thresholds, validation-based pruning, and deterministic rank reduction before the final solve. Experiments report the full tree-size and forest-size paths, not only the selected model.

References

[1] Susan Athey, Julie Tibshirani, and Stefan Wager. Generalized random forests. *The Annals of Statistics*, 47(2):1148–1178, 2019.

[2] Leemon Baird et al. Residual algorithms: Reinforcement learning with function approximation. In *Proceedings of the twelfth international conference on machine learning*, pages 30–37, 1995.

[3] Dimitri P Bertsekas. Approximate policy iteration: A survey and some new methods. *Journal of Control Theory and Applications*, 9(3):310–335, 2011.

[4] Leo Breiman. Random forests. *Machine learning*, 45:5–32, 2001.

[5] John B Conway. *A course in functional analysis*, volume 96. Springer Science & Business Media, 1994.

[6] Damien Ernst, Pierre Geurts, and Louis Wehenkel. Tree-based batch mode reinforcement learning. *Journal of Machine Learning Research*, 6, 2005.

[7] Wassily Hoeffding. Probability inequalities for sums of bounded random variables. *Journal of the American Statistical Association*, 58(301):13–30, 1963.

[8] Roger A Horn and Charles R Johnson. *Matrix Analysis*. Cambridge University Press, 2 edition, 2012.

[9] Michail G Lagoudakis and Ronald Parr. Least-squares policy iteration. *Journal of machine learning research*, 4(Dec):1107–1149, 2003.

[10] Rémi Munos and Csaba Szepesvári. Finite-time bounds for fitted value iteration. *Journal of Machine Learning Research*, 9(5), 2008.

[11] Erwan Scornet. Random forests and kernel methods. *IEEE Transactions on Information Theory*, 62(3):1485–1500, 2016.

[12] Joel A Tropp. User-friendly tail bounds for sums of random matrices. *Foundations of Computational Mathematics*, 12(4):389–434, 2012.

[13] John Tsitsiklis and Benjamin Van Roy. Analysis of temporal-diffference learning with function approximation. *Advances in neural information processing systems*, 9, 1996.

[14] Masatoshi Uehara, Masaaki Imaizumi, Nan Jiang, Nathan Kallus, Wen Sun, and Tengyang Xie. Finite sample analysis of minimax offline reinforcement learning: Completeness, fast rates and first-order efficiency. *arXiv preprint arXiv:2102.02981*, 2021.

[15] Benjamin Van Roy. Performance loss bounds for approximate value iteration with state aggregation. *Mathematics of Operations Research*, 31(2):234–244, 2006.

[16] Stefan Wager and Susan Athey. Estimation and inference of heterogeneous treatment effects using random forests. *Journal of the American Statistical Association*, 113(523):1228–1242, 2018.

[17] Martin J Wainwright. *High-dimensional statistics: A non-asymptotic viewpoint*, volume 48. Cambridge university press, 2019.

[18] Tengyang Xie and Nan Jiang. Q* approximation schemes for batch reinforcement learning: A theoretical comparison. In *Conference on Uncertainty in Artificial Intelligence*, pages 550–559. PMLR, 2020.

[19] Tengyang Xie, Ching-An Cheng, Nan Jiang, Paul Mineiro, and Alekh Agarwal. Bellman-consistent pessimism for offline reinforcement learning. *Advances in neural information processing systems*, 34:6683–6694, 2021.

A Proofs

Standard facts used below. We use the Hilbert-space projection theorem and Banach fixed-point theorem [5], Weyl's singular-value perturbation inequality [8], Hoeffding's inequality for bounded random variables [7], matrix Bernstein [12], and standard vector concentration for bounded random vectors [17].

Proof of Lemma 1. For any bounded f, g, rewards cancel and

$$T^\pi f(X) - T^\pi g(X) = \gamma(P_\pi(f-g))(X).$$

By Jensen's inequality,

$$\{P_\pi(f-g)(X)\}^2 \le P_\pi\{(f-g)^2\}(X).$$

Taking $X \sim \mu_\pi$ and using stationarity of μ_π for the target-policy chain gives

$$\|P_\pi(f-g)\|_{L^2(\mu_\pi)}^2 \le \mathbb{E}_{\mu_\pi}[(f-g)^2(X_+^\pi)] = \|f-g\|_{L^2(\mu_\pi)}^2.$$

Multiplying by γ^2 proves the claim. □

Proof of Proposition 1. Because $\mathcal{V}_\psi$ is finite dimensional, it is a closed subspace of $L^2(\mu)$. The orthogonal projection $\Pi_\psi^\mu : L^2(\mu) \to \mathcal{V}_\psi$ is therefore well defined, maps every function into $\mathcal{V}_\psi$, and is nonexpansive. For $f, g \in \mathcal{V}_\psi$,

$$\|\Pi_\psi^\mu T^\pi f - \Pi_\psi^\mu T^\pi g\|_{L^2(\mu)} \le \|T^\pi f - T^\pi g\|_{L^2(\mu)} \le \gamma \|f-g\|_{L^2(\mu)}.$$

Thus $\Pi_\psi^\mu T^\pi$ is a contraction on the complete metric space $\mathcal{V}_\psi$, so Banach's theorem gives a unique fixed function $Q_\psi^\mu \in \mathcal{V}_\psi$.

If $Q(x) = \psi(x)^\top \theta$, the projected fixed-point equation is equivalent to the residual $Q - T^\pi Q$ being orthogonal to every element of $\mathcal{V}_\psi$. Since the coordinates of ψ span $\mathcal{V}_\psi$, this is equivalent to

$$\mathbb{E}_\mu[\psi(X)\{Q(X) - R - \gamma Q(X_+^\pi)\}] = 0,$$

which is exactly $A_\psi^\mu \theta = b_\psi^\mu$. The fixed function is unique by contraction. The coefficient vector is unique whenever A_ψ^μ is nonsingular, which is enforced after rank reduction in Assumption 3. □

Proof of Theorem 1. Using Proposition 1, $Q^\pi = T^\pi Q^\pi$, nonexpansiveness of Π_ψ^μ, and Assumption 2,

$$\begin{aligned}
\|Q_\psi^\mu - Q^\pi\|_{L^2(\mu)} &= \|\Pi_\psi^\mu T^\pi Q_\psi^\mu - T^\pi Q^\pi\|_{L^2(\mu)} \\
&\le \|\Pi_\psi^\mu (T^\pi Q_\psi^\mu - T^\pi Q^\pi)\|_{L^2(\mu)} + \|(I - \Pi_\psi^\mu) Q^\pi\|_{L^2(\mu)} \\
&\le \gamma \|Q_\psi^\mu - Q^\pi\|_{L^2(\mu)} + \|(I - \Pi_\psi^\mu) Q^\pi\|_{L^2(\mu)}.
\end{aligned}$$

Rearranging proves the bound. For one-hot tree features, $\Pi_{\mathcal{T}}^\mu Q^\pi$ is the conditional mean $\mathbb{E}_\mu[Q^\pi(X) \mid \phi_{\mathcal{T}}(X)]$. The displayed variance identity is therefore the conditional-variance decomposition

$$\mathbb{E}_\mu\big[\{Q^\pi(X) - \mathbb{E}_\mu[Q^\pi(X) \mid \phi_{\mathcal{T}}(X)]\}^2\big] = \mathbb{E}_\mu[\mathrm{Var}\{Q^\pi(X) \mid \phi_{\mathcal{T}}(X)\}].$$

□

Proof of Proposition 2. Orthogonal projection onto a larger closed subspace cannot have larger least-squares approximation error. Since $\mathcal{V}_{\psi_1} \subseteq \mathcal{V}_{\psi_2}$, the best approximation to Q^π in $\mathcal{V}_{\psi_2}$ is at least as good as the best approximation in $\mathcal{V}_{\psi_1}$. □

Proof of Theorem 2. Write $A = A_\psi^\mu$ and $b = b_\psi^\mu$. Subtract $A\theta = b$ from $\widehat{A}\hat{\theta} = \widehat{b}$:

$$\widehat{A}(\hat{\theta} - \theta) = \widehat{b} - b - (\widehat{A} - A)\theta.$$

Weyl's inequality gives $\sigma_{\min}(\widehat{A}) \ge \kappa - \varepsilon_A$, hence

$$\|\hat{\theta} - \theta\|_2 \le \frac{\varepsilon_b + \varepsilon_A \|\theta\|_2}{\kappa - \varepsilon_A}.$$

The value-function bound follows from $\|\psi^\top(\hat{\theta} - \theta)\|_{L^2(\mu)} \le \sup_x \|\psi(x)\|_2 \|\hat{\theta} - \theta\|_2$. □

Proof of Proposition 3. Condition on the fixed retained feature map. Define

$$Y_i^A = w_i \psi_i (\psi_i - \gamma \bar{\psi}_i^+)^\top, \qquad Y_i^b = w_i \psi_i R_i.$$

Then $\mathbb{E}_\nu Y_i^A = A_\psi^\mu$ and $\mathbb{E}_\nu Y_i^b = b_\psi^\mu$. Moreover

$$\|Y_i^A\|_{\mathrm{op}} \le W K^2 (1+\gamma), \qquad \|Y_i^b\|_2 \le W K R_{\max}.$$

Matrix Bernstein and vector Bernstein imply that, with probability at least $1 - 2\delta/3$,

$$\|\widetilde{A}_\psi - A_\psi^\mu\|_{\mathrm{op}} \le C_0 W K^2 (1+\gamma) \left\{ \sqrt{\frac{d + \log(6/\delta)}{n}} + \frac{\log(6d/\delta)}{n} \right\},$$

and

$$\|\widetilde{b}_\psi - b^\mu_\psi\|_2 \le C_0 W K R_{\max} \sqrt{\frac{d + \log(6/\delta)}{n}}.$$

Since $0 \le w_i \le W$ and $\mathbb{E}_\nu w_i = 1$, Hoeffding's inequality gives

$$\Pr\{|D_n - 1| > t\} \le 2\exp\{-2nt^2/W^2\}.$$

With $t_n = W\sqrt{\log(6/\delta)/(2n)}$, this event has probability at least $1 - \delta/3$. In particular,

$$\Pr\{|D_n - 1| > 1/2\} \le 2\exp\{-n/(2W^2)\},$$

and the assumed lower bound on n ensures $t_n \le 1/2$. On the event $|D_n - 1| \le t_n \le 1/2$,

$$\|\widehat{A}_\psi - A^\mu_\psi\|_{\mathrm{op}} = \left\|\frac{\widetilde{A}_\psi}{D_n} - A^\mu_\psi\right\|_{\mathrm{op}} \le 2\|\widetilde{A}_\psi - A^\mu_\psi\|_{\mathrm{op}} + 2|D_n - 1|\,\|A^\mu_\psi\|_{\mathrm{op}}.$$

The same argument gives

$$\|\widehat{b}_\psi - b^\mu_\psi\|_2 \le 2\|\widetilde{b}_\psi - b^\mu_\psi\|_2 + 2|D_n - 1|\,\|b^\mu_\psi\|_2.$$

Finally, $\|A^\mu_\psi\|_{\mathrm{op}} \le K^2(1+\gamma)$ and $\|b^\mu_\psi\|_2 \le K R_{\max}$. Absorbing constants into a universal constant C proves the stated inequalities. □

Proof of Corollary 1. For one-hot tree features, the normal equations are

$$D_\mu(I - \gamma P^\mu)q = D_\mu r^\mu,$$

which reduces to $q = r^\mu + \gamma P^\mu q$ on leaves with positive μ-mass. The empirical equation is the same with $(\widehat{r}^\mu, \widehat{P}^\mu)$, where

$$\widehat{r}^\mu_\ell = \frac{n^{-1}\sum_i w_i \mathbb{1}\{\phi_{\mathcal{T}}(X_i) = \ell\} R_i}{\hat{\mu}_\ell}, \qquad \widehat{P}^\mu_{\ell m} = \frac{n^{-1}\sum_i w_i \mathbb{1}\{\phi_{\mathcal{T}}(X_i) = \ell\} p_{\mathcal{T},m}(S'_i)}{\hat{\mu}_\ell}.$$

For illustration, let

$$N^r_\ell = \mathbb{E}_\nu[w \mathbb{1}\{\phi_{\mathcal{T}}(X) = \ell\} R], \qquad \hat{N}^r_\ell = n^{-1}\sum_i w_i \mathbb{1}\{\phi_{\mathcal{T}}(X_i) = \ell\} R_i.$$

On $\hat{\mu}_\ell \ge \mu_\ell/2$,

$$|\widehat{r}^\mu_\ell - r^\mu_\ell| = \left|\frac{\hat{N}^r_\ell}{\hat{\mu}_\ell} - \frac{N^r_\ell}{\mu_\ell}\right| \le \frac{2|\hat{N}^r_\ell - N^r_\ell|}{\mu_\ell} + \frac{2R_{\max}|\hat{\mu}_\ell - \mu_\ell|}{\mu_\ell}.$$

The same ratio calculation applies entrywise to $\widehat{P}^\mu_{\ell m}$, with $R_{\max}$ replaced by 1. Thus the displayed assumptions $\|\widehat{r}^\mu - r^\mu\|_\infty \le \varepsilon_r$ and $\|\widehat{P}^\mu - P^\mu\|_\infty \le \varepsilon_P$ are the corresponding self-normalized leaf-moment errors.

Subtracting the two Bellman equations and using $\|(I - \gamma \widehat{P}^\mu)^{-1}\|_\infty \le (1-\gamma)^{-1}$ and $\|q\|_\infty \le R_{\max}/(1-\gamma)$ gives the displayed bound. The leaf-mass condition follows because

$$\Pr\{|\hat{\mu}_\ell - \mu_\ell| > \mu_{\min}/2\} \le 2\exp\{-n\mu^2_{\min}/(2W^2)\}$$

for each leaf by Hoeffding, and a union bound over L leaves gives the claimed sample-size scaling. □

Proof of Proposition 4. Apply the same perturbation argument as in Theorem 2 with $(A, b) = (A^w, b^w)$ and $(\widehat{A}, \widehat{b}) = (A^{\hat{w}}, b^{\hat{w}})$. □

Proof of Proposition 5. Condition on the candidate set and on all non-validation data used to compute the provisional residual functions. For a fixed candidate, $w_i \delta^2_{\psi,i} \in [0, W B^2_\delta]$. Hoeffding's inequality and a union bound over $\mathfrak{P}$ imply a uniform deviation bound of

$$W B^2_\delta \sqrt{\frac{\log(2|\mathfrak{P}|/\delta)}{2n_{\mathrm{val}}}}.$$

Call this deviation ε_{val}. Let $\psi^\star$ minimize the population penalized risk over $\mathfrak{P}$. On the uniform event,

$$\begin{aligned} L_\mu(\hat{\psi}) + \lambda \dim(\operatorname{span} \hat{\psi}) &\le \widehat{L}_{\text{val}}(\hat{\psi}) + \lambda \dim(\operatorname{span} \hat{\psi}) + \varepsilon_{\text{val}} \\ &\le \widehat{L}_{\text{val}}(\psi^\star) + \lambda \dim(\operatorname{span} \psi^\star) + \varepsilon_{\text{val}} \\ &\le L_\mu(\psi^\star) + \lambda \dim(\operatorname{span} \psi^\star) + 2\varepsilon_{\text{val}}. \end{aligned}$$

This proves the result. If the empirical score is divided by the validation denominator D_{val}, the same argument applies on the event $|D_{\text{val}} - 1| \le 1/2$, with the displayed deviation multiplied by a universal constant. □

B Implementation Details

Candidate splits. For continuous variables, candidate thresholds are empirical quantiles inside each leaf after enforcing minimum raw count, weighted mass, and ESS. For categorical variables and finite actions, the implementation considers binary subsets when arity is small and one-vs-rest splits otherwise. The action is treated as part of the state-action feature vector.

Rank reduction. For the theorem-clean estimator, forest features are reduced on an independent rank fold by QR with column pivoting or SVD using a fixed tolerance. The retained feature map is then held fixed for final estimation. A practical implementation may run the same QR/SVD on the estimation design because it uses no rewards, but this should be reported as a data-dependent numerical preprocessing step. A ridge fallback is allowed only as a diagnostic because it changes the projected equation.

Residual scoring. The residual score in (3.1) uses held-out weighted transitions. A candidate's provisional value is estimated on the growth-fit fold, then residuals are computed on growth-score or validation transitions by assigning current and next state-action pairs to candidate features and weighting by $\hat{w}_i$.

C Experiment Templates

Diagnostic MDP template. Use a two-dimensional state $S = (S_1, S_2) \in [-1, 1]^2$ and two actions. Let one coordinate primarily determine immediate rewards and the other determine which high- or low-value region the target policy reaches next. Display true Q^π, CART/RF supervised partitions or weights, Tree-FQE/RF-FQE behavior, BAT partitions, and BAF forest weights at matched feature budgets.

Navigation template. Use a grid-auditable continuous navigation task with five actions. Compute reference Q^π by dynamic programming on a fine grid. Draw offline transitions from behavior policies with increasing mismatch from the target stationary distribution. Report errors under both the target and behavior distributions to show when ordinary supervised criteria optimize the wrong geometry.